Dressed to Kill

How Arsenic Green Took Over the 1800s

15 MINUTE HISTORY · NARRATION SCRIPT · 1,648 WORDS · ABOUT 9:55

> ***Driving question:*** *How did a poison end up on the walls, on the dresses, and on the candy? And why did it take so long to get it off?*

Underlined = vocabulary card on screen · *italic* = spoken with emphasis · timestamps are estimates until the voice is recorded

0:00 The Library Book

November 1874. Lansing, Michigan.

This book goes out in the mail to 100 public libraries. Inside, just page after page of wallpaper. 86 samples, all of it on sale in ordinary stores.

Now look at the letter in the front. Because of "the poisonous nature of the paper of which this book is made," it says, maybe don't let children handle it.

Every page has *arsenic* in it.

And the very first sample came from the house where a girl, about ten years old, had died that year.

Only four or five copies survive. Librarians handle them with gloves. When one was scanned in 2018, the person scanning it wore a protective suit.

By 1874, everybody knew arsenic was poison. So here's the question I want to answer today: How did a poison end up on the walls, on the dresses, and on the candy? And why did it take so long to get it off?

It starts with a Swedish chemist, a beautiful new color, and a warning.

1:10 The Perfect Green

1775. Sweden. A chemist named Carl Wilhelm Scheele mixes copper with arsenic and gets something remarkable: a bright, clean green. Brighter than any plant dye. And cheap.

It's called **Scheele's Green**. And Scheele knew exactly what was in it. A couple of years later he wrote that people using it ought to be warned it was poisonous.

Not a great sign.

In 1814, German chemists made an even brighter version, later called **Paris green**. For this story they're the same thing, and I'm going to call them what one Victorian poet did: *arsenic green*.

So why did everyone want it? Timing. Factories were filling cities with smoke and soot. Green was the color of the countryside, of everything the city wasn't. And with new printing machines, ordinary families could afford it.

In 1834, Britain made about 1.2 million rolls of wallpaper. Forty years later? About 32 million.

So arsenic green went everywhere. On parlor walls. On ball gowns. In the fake leaves women pinned in their hair. On candy wrappers, children's toys, even book covers.

All of it, arsenic.

2:16 Dressed to Kill

The most dangerous place to wear it? Possibly a dance.

The 1860s fashion was enormous skirts of thin fabric, often dyed bright green, with green leaves and flowers pinned in the hair.

In 1862, a chemist tested one of those hair wreaths. It held enough arsenic, he said, "to poison twenty persons."

The British Medical Journal did the math on a whole dress. A lady at a ball, it wrote, "carries in her skirts poison enough to slay the whole of the admirers she may meet with in half a dozen ball-rooms."

Which is one way to keep admirers at a distance.

The same month, the comic magazine *Punch* ran this. "The Arsenic Waltz." A gentleman bows to a lady and asks her to dance. Now look closer. They're both skeletons.

But here's the twist. Most of the women wearing those dresses were fine. One evening in a green gown usually wasn't enough to hurt you. The people in real danger were the ones who *made* them, breathing that green dust all day, every day.

Hold onto that. We're about to meet one of them.

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If you're enjoying this, hit subscribe. Unlike a Victorian ball gown, it's completely safe to touch.

3:30 The Walls *(heavy chapter: no jokes, slower voice)*

London, 1861. Matilda Scheurer is 19 years old. She makes **artificial flowers**, dusting fake leaves with green powder, in a workshop with about a hundred other young women. They tie thin cloth over their mouths. It doesn't help much.

After a year and a half, she gets sick. She throws up something green. In November 1861, she dies. A doctor finds arsenic in her lungs and her liver. Her sister had died the same way.

The **inquest**, the official investigation into her death, calls it an accident. Caused, it says, by the arsenic "used in her employment."

A few months later, on the other side of London, a bricklayer's family named Turner is losing children, one after another. The doctors call it diphtheria, a throat infection. Then the youngest, three-year-old Ann Amelia, gets sick too.

The surgeon who comes to see her, Thomas Orton, notices the walls. Green paper. Damp. The color came off on your hand "like paint."

A chemist tests it. Arsenic. Four children in about two months.

The jury at the inquest hears all of this, and decides the children died of natural causes.

And that was the pattern. People got sick at home. They got better when they went away. They came home, and got sick again. And very often, nobody thought to look at the walls.

4:51 Witch Fever

So if doctors were finding arsenic in the walls, why didn't it stop? Three reasons.

One: the evidence was messy. Plenty of people lived with green wallpaper and felt fine. And arsenic poisoning looks like a lot of other illnesses. So every case could be argued.

Two: doctors couldn't agree on *how* a wall could poison you. One theory blamed a garlic-smelling gas from mold on damp walls. That gas is real, but modern tests suggest it's pretty weak. The likelier culprit was simpler. Arsenic green flaked off as dust. Every time someone brushed past the wall. Every time a child touched the paper, then put a finger in their mouth. That means the poison wasn't just sitting in the walls. It was coming off them.

You might have heard that wallpaper killed Napoleon. His really did have arsenic in it. But it almost certainly didn't kill him. Stomach cancer did.

Three: money. One English wallpaper maker said he went through two tons of arsenic a week.

Which brings us to William Morris, probably the most famous designer of his day. His family's fortune came from an arsenic mine, one of the biggest in the world, and from 1871 to 1875 he sat on its board. His early wallpapers used arsenic green.

In 1885, Morris wrote to a colleague: "As to the arsenic scare a greater folly it is hardly possible to imagine: the doctors were bitten as people were bitten by the witch-fever."

Witch fever. In other words: a panic. His own theory? The real problem was the toilets.

To be fair to Morris, his company had stopped using arsenic years earlier, and the science really was disputed. To his critics, it's a man whose fortune came out of an arsenic mine, calling the danger a panic.

6:40 Shall We Poison Our Homes?

Meanwhile, in America, the same green was on the same walls.

Massachusetts, 1872. A Boston doctor named Frank Draper writes a report for the state's **board of health**, a government office that tracks disease. His first line: "Everybody knows that arsenic is a deadly poison."

Then he shows where it's hiding. Bedroom walls. Children's paint boxes. Candy wrappers in the window of one of Boston's best shops. He even bound real samples of the poisoned paper into his report.

So, ban it? Some European governments were already restricting these colors. Draper didn't think it would work here. That would be difficult, he wrote, "under a government organized like our own. The rights of individuals and of industrial pursuits are deemed too sacred."

Translation: he didn't think Americans would let the government tell a business what it couldn't sell. That's **laissez-faire**, French for "leave it

alone." The government stays out, and buyers look out for themselves. Stop buying it, Draper said, and they'll stop making it.

Two years later, in Michigan, a chemistry professor named Robert Kedzie decided people needed to see it for themselves.

He collected cases. A nine-year-old girl named Emma got sick in her green bedroom, and got better when she was moved out.

And in a little town called Spring Port, the girl from the first page. Her family's walls were green in nearly every room.

So Kedzie went shopping. He gathered wallpaper from ordinary stores and tested every sheet himself. All 86 had arsenic. Then he bound them into a book, and mailed it to the libraries. This book.

His headline was a question: "Shall we poison our homes?"

8:19 Arsenic-Free

So how did it end? In Britain, not with a law.

Doctors kept writing, newspapers kept printing, and shoppers got scared. By the 1880s, "arsenic-free" was something you put in the ad.

Other countries did pass bans. Sweden in 1876. Germany in the 1880s. But Britain never banned arsenic wallpaper. It didn't have to. By about 1900, nobody wanted it, so nobody made it.

In America, Draper's "leave it alone" idea got a long test. Massachusetts finally set a legal limit on arsenic in wallpaper and cloth in 1900. That's 28 years after his report.

But the green didn't disappear. It moved. Back in the 1860s and 70s, a beetle was eating its way across America's potato fields. Farmers found that Paris green killed it. So the same poison Americans were tearing off their walls, they were also dusting onto their potatoes. It became one of the first big insecticides in the country.

9:16 Back to the Book

So let's go back to the question. How did a poison end up on the walls? And why did it take so long to get it off?

It got there because it was beautiful, and cheap. It stayed because the proof was messy, the money was big, and nobody had to prove it was safe. It was up to the sick to prove it wasn't. A family in Limehouse couldn't. A jury looked at four children and green walls, and called it natural.

In Britain, it was shoppers who finally got it off the walls, not lawmakers.

Kedzie's book is still out there, sealed in plastic. Still green.

The green never changed. What changed was who had to prove it was safe.

William Morris called it witch fever. But this time, the witch was real.

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